4.4. 对 $\varphi(\alpha)=A\alpha$ $\varphi^2(\alpha)=A(A\alpha)=A^2\alpha=A\alpha=\varphi(\alpha)$

$\therefore \varphi^2=\varphi$ 即φ是幂等变化

5、$\because \varphi^{m-1}(\alpha)\neq 0$ $\therefore \varphi^n(\alpha)\neq 0$ $(0\leq n\leq m-1)$

若 α, $\varphi(\alpha)$, … $\varphi^{m-1}(\alpha)$ 线性相关

则有在 $\varphi^{m-1}(\alpha)=k_1\alpha+k_2\varphi(\alpha)+\cdots+k_{m-1}\varphi^{m-2}(\alpha)$

$\therefore \varphi^m(\alpha)=\varphi(k_1\alpha+k_2\varphi(\alpha)+\cdots+k_{m-1}\varphi^{m-2}(\alpha))$

$=k_1\varphi(\alpha)+k_2\varphi^2(\alpha)+\cdots+k_{m-1}\varphi^{m-1}(\alpha)\neq 0$

导出矛盾，故线性无关

6、$\because a_n\neq 0$ $\therefore \frac{1}{a_n}(\varphi^n+a_1\varphi^{n-1}+\cdots+a_{n-1}\varphi)=I_V$

$\therefore \varphi(\frac{1}{a_n}(\varphi^{\overset{n-1}{\cancel{n-2}}}+\frac{1}{a_1}\varphi^{n-\cancel{3}}+\cdots+a_{n-2}\varphi\overset{+a_{n-1}}{)}=\frac{1}{a_n}(\varphi^{n-1}+\frac{1}{a_1}\varphi^{n-2}+\cdots+a_{n-2}\varphi+a_{n-1})\varphi=I_V$

$\therefore \varphi^{-1}=\frac{1}{a_n}(\varphi^{n-1}+\frac{1}{a_1}\varphi^{n-2}+\cdots+a_{n-1})$

$\therefore \varphi$是V上的自同构

7、$\Leftarrow$ 若 φ 将基 $\{e_1e_2\ldots e_n\}$ 变为基 $\{f_1,f_2\ldots f_n\}$

易知存在变化 ψ 将基 $\{f_1\ldots f_n\}$ 变为 $\{e_1\ldots e_n\}$

$\therefore \varphi\circ\psi=I_V$ $\therefore \varphi$ 可逆

$\Rightarrow$ 若φ可逆，设φ: $\varphi(\alpha)=A\alpha$，则 A可逆，$\therefore AA^{-1}=I_n$

设 V的一组基为 $\{e_1,e_2\ldots \cancel{A}e_n\}$

则对 $k_1Ae_1+k_2Ae_2+\cdots+k_3Ae_n=0 \Leftrightarrow \cancel{A}\ A^{-1}(k_1Ae_1+k_2Ae_2+\cdots+k_nAe_n)=0$

$\therefore k_1e_1+k_2e_2+\cdots+k_ne_n=0$

$\therefore k_1=k_2=\cdots=k_n=0$

$\therefore \{\varphi(e_1),\ldots\varphi(e_n)\}$ 线性无关 是V的一组基(数量为n)

$\therefore \varphi$将V的基变为基

$4.\overset{3}{4}.1$ $A=\begin{pmatrix} 0 & & & & \\ n-1 & 0 & & & \\ & n-2 & 0 & & \\ & & \ddots & \ddots & \\ & & & 1 & 0 \end{pmatrix}$

2.(1) $\begin{pmatrix} 2 & 3 & 2 & 1 \\ 1 & 3 & 0 & -1 \\ -1 & 5 & 1 & 2 \\ 2 & 2 & 1 & 1 \end{pmatrix}$ (2) $\begin{pmatrix} 1 \\ -1 \\ 2 \\ 1 \end{pmatrix}$ (1) $\begin{pmatrix} 2 & 2 & 1 & 1 \\ -1 & 5 & 1 & 2 \\ 1 & 3 & 0 & -1 \\ 2 & 3 & 2 & 1 \end{pmatrix}$ 设在 $\{e_1,e_2,e_3,e_4\}$上 $\varphi(\alpha)=A\alpha$

(1) $(e_1,e_2,e_3,e_4)=\begin{pmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{pmatrix}(e_4,e_3,e_2,e_1)$

则在 $\{e_4,e_3,e_2,e_1\}$上，$\varphi\left[\begin{pmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{pmatrix}\alpha\right]=A\begin{pmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{pmatrix}\alpha$ $\therefore \varphi(\alpha)=\begin{pmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{pmatrix}A\begin{pmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{pmatrix}\alpha$

$\therefore \varphi$在$\{e_4,e_3,e_2,e_1\}$上表示的矩阵为 $\begin{pmatrix} 2 & 2 & 1 & 1 \\ -1 & 5 & 1 & 2 \\ 1 & 3 & 0 & -1 \\ 2 & 3 & 2 & 1 \end{pmatrix}$

(2) $(e_1,e_2,e_3,e_4)=\begin{pmatrix} 1 & & & \\ -1 & 1 & & \\ & -1 & 1 & \\ & & -1 & 1 \end{pmatrix}(e_1,\ e_1+e_2,\ e_1+e_2+e_3,\ e_1+e_2+e_3+e_4)$

Recording start from here

则在 $\{e_1, e_1+e_2, e_1+e_2+e_3, e_1+e_2+e_3+e_4\}$ 上，

$\varphi\left[\begin{pmatrix}1&&&\\-1&1&&\\&-1&1&\\&&-1&1\end{pmatrix}\alpha\right] = A\begin{pmatrix}1&&&\\-1&1&&\\&-1&1&\\&&-1&1\end{pmatrix}\alpha$

$\therefore \varphi(\alpha) = \begin{pmatrix}1&&&\\-1&1&&\\&-1&1&\\&&-1&1\end{pmatrix}^{-1} A \begin{pmatrix}1&&&\\-1&1&&\\&-1&1&\\&&-1&1\end{pmatrix}\alpha$

∴ 表示矩阵为 $\begin{pmatrix}2&4&4&5\\-3&-4&-6&-4\\1&1&9&1\\1&2&4&6\end{pmatrix}$

3、设 $\alpha =$ ~~$(r\cos\theta,$~~ $(r\cos\beta, r\sin\beta)^T$

$\therefore \varphi(\alpha) = (r\cos(\beta+\theta), r\sin(\beta+\theta))^T = (r\cos\beta\cos\theta - r\sin\beta\sin\theta, r\sin\beta\cos\theta + r\sin\theta\cos\beta)^T$

$= \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\alpha$

∴ 表示的矩阵为 $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$

4、设在第一组基下 $(\psi\varphi^{-2}+2\varphi+I_V)(\alpha) = (BA^{-2}+2A+I_n)\alpha$

则在第二组基下 $(\psi\varphi^{-2}+2\varphi+I_V)(P\alpha) = (BA^{-2}+2A+I_n)P\alpha$

$\therefore (\psi\varphi^{-2}+2\varphi+I_V)\ \alpha = P^{-1}(BA^{-2}+2A+I_n)P$

∴ 表示矩阵为 $P^{-1}(BA^{-2}+2A+I_n)P$

5、$\varphi: V\to U$ ~~可看为~~ 与一个 $m\times n$ 的矩阵等价

∴ $\mathcal{L}(V, U)$ ~~可看为~~ 与 $m\times n$ 的矩阵全体等价

∴ $\mathcal{L}(V, U)$ 是 mn ~~m.n~~ 维线性空间。

6、设 ~~α~~ A, B 为 n 阶矩阵，且 a, A, B 相似，

设 α 为 n 阶向量，$\{\lambda_n\}$ 满足 $(\lambda I_n - A)\alpha = 0$ （α 非零）

则 $|\lambda I_n - A| = \begin{pmatrix}\lambda-a_{11} & -a_{12} & & \\ -a_{21} & \lambda-a_{22} & & \\ & & \ddots & \\ & & & \lambda_n-a_{nn}\end{pmatrix} = k_1\lambda^n + k_2\lambda^{n-1} + \cdots + K_n\lambda^n + K_{n+1}$ ①

$\because \alpha$ 非零 $\therefore |\lambda_n I_n - A| = 0$ $\therefore \lambda_n$ 是 $|\lambda I_n - A| = 0$ 的解

由①式 ~~$tr(A)=$~~ 及 $|\lambda I_n - A| = (\lambda-\lambda_1)(\lambda-\lambda_2)\cdots(\lambda-\lambda_n) = 0$ 得

$tr(A) = a_{11}+a_{22}+\cdots+a_{nn} = -k_2 = \lambda_1+\cdots+\lambda_n$

$|A| = (-1)^n a_n = \lambda_1\lambda_2\cdots\lambda_n$

而对 $|\lambda I_n - B| = |P^{-1}(\lambda I_n - B)P| = |\lambda I_n - A|$ 故 A 与 B 的特征值相同

故 A 与 B 迹与行列式不变

7、~~AB~~ $BA = A^{-1}(AB)A = BA$ ∴ AB 与 BA 相似

8、若 A 只与自己相似，则 $A = P^{-1}AP$，$PA = AP$ ∴ A 可与任一可逆阵交换

∴ 由 2 中结论知 $A = cI_n$

9、设 $A = P^{-1}BP$ $C = Q^{-1}DQ$ 则 $\begin{pmatrix}A&0\\0&C\end{pmatrix} = \begin{pmatrix}P^{-1}&\\&Q^{-1}\end{pmatrix}\begin{pmatrix}B&\\&D\end{pmatrix}\begin{pmatrix}P&\\&Q\end{pmatrix}\left[\begin{pmatrix}P^{-1}&\\&Q^{-1}\end{pmatrix}\begin{pmatrix}P&\\&Q\end{pmatrix} = \begin{pmatrix}I_n&\\&I_m\end{pmatrix} = I\right]$

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10、设得到的矩阵为A'

(1) $A' = P_{ij}^{-1} A P_{ij}$

(2) $A' = P_i(c) A P_i(c)^{-1}$

(3) $A' = T_{ij}(c) A P_{ji}^{-1}(c)$

11、对任一相似变化 ~~$A' = P^{-1}AP$~~ $\sigma_p(A) = P^{-1}AP$

∵ P 可逆 ∴ P 可折为若干初等矩阵的乘积

∴ $\sigma_p(A) = (P_n^{-1} \cdots P_1^{-1}) A (P_1 \cdots P_n) = (P_n^{-1}(P_{n-1}^{-1}(\cdots (P_1^{-1} A P_1) \cdots P_{n-1}) P_n)$

$= \sigma_{P_n} \circ \sigma_{P_{n-1}} \circ \cdots \circ \sigma_{P_1}(A)$

得证

12、~~(1) 设~~ 设得到的矩阵为A'

(1) A' ~~$= (P_{\cdots}$~~ 相当于进行~~若干~~m次行变交换再进行m次列变化，且行，列变化对象相同

∴ $A' = (P_n^{-1} \cdots P_1^{-1}) A (P_1 \cdots P_n) = P^{-1}AP$ （P_n为第一类初等矩阵）

(2) $A' = \begin{pmatrix} I & & & \\ & I \ddots & & \\ & & m & \\ & & & I \end{pmatrix} A \begin{pmatrix} I & & & \\ & I \ddots & & \\ & & m & \\ & & & I \end{pmatrix}^{-1}$ $A = \begin{pmatrix} I & & & \\ & \ddots & I & \\ & I & \ddots & \\ & & & I \end{pmatrix}^{-1} A \begin{pmatrix} I & & & \\ & \ddots & I & \\ & I & \ddots & \\ & & & I \end{pmatrix}$

(3) $A' = \begin{pmatrix} I & & & \\ & I & & \\ & m & I & \\ & & & I \end{pmatrix} A \begin{pmatrix} I & & & \\ & I & & \\ & m & I & \\ & & & I \end{pmatrix}^{-1}$

13、设 φ 在基$\{e_1 \ldots e_n\}$的表示矩阵为A， $B = P^{-1}AP$

则 φ 在基 $\{P^{-1}e_1 \ldots P^{-1}e_n\}$ 的表示矩阵为 $P^{-1}AP = B$，得证

4.4.1 设 $A = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ ∴ $\alpha_1 \neq k\alpha_2$，$2\alpha_1 + \frac{3}{2}\alpha_2 = \alpha_3$ $\alpha_1 + 2\alpha_2 = \alpha_4$

∴ $r(A) = 2$ 且 α_1, α_2是$\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$的一个极大线性无关组

∴ ~~$\operatorname{Im}\varphi = k_1\alpha_1 + k_2\alpha_2$ $\ker\varphi = k_1\alpha_3 + k_2\alpha_4$~~

∴ $\operatorname{Im}\varphi = k_1(e_1 - e_2 + e_3 + 2e_4) + k_2(2e_2 + 2e_3 - 2e_4)$

求 $Ax = 0$，得其一个基础解系为 $k_1(-4, -3, 2, 0) + k_2(1, 2, 0, -1)$

∴ $\ker\varphi = k_1(-4e_1 - 3e_2 - 2e_3) + k_2(e_1 + 2e_2 - e_4)$

2、对 $A \in V$ $A = A_1 + A_2$ $(A_1 \in V_1, A_2 \in V_2)$

∴ $\varphi^2(A) = \varphi \circ \varphi(A) = \varphi(A_1) = \varphi(A)$ ∴ $\varphi^2 = \varphi$

∴ $\operatorname{Im}\varphi = V_1$ $\ker\varphi = V_2$

表示矩阵 $B = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$

3、设对$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\varphi(A) = \begin{pmatrix} a+c & b+d \\ a+c & b+d \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2(a+c) & b+d \\ 2(a+c) & b+d \end{pmatrix} \Rightarrow$ A

∴ $\ker\varphi = k_1\begin{pmatrix} a & 0 \\ -a & 0 \end{pmatrix} + k_2\begin{pmatrix} 0 & b \\ 0 & -b \end{pmatrix}$

∴ $\dim\ker\varphi = 2$ ~~故~~ φ 的秩为 $4 - 2 = 2$

Recording start from here

4、由秩的性质 $\varphi(A+B)=tr(A+B)=trA+trB=\varphi(A)+\varphi(B)$

$\varphi(KA)=tr(KA)=K\,tr A=K\varphi(A)$

$\because \varphi$ 是 $V\to K$ 的线性映射

$\therefore \dim\ker\varphi=n-1$

$\ker\varphi$ 基为 $\begin{pmatrix}1&&&\\&0&&\\&&0&\\&&&\ddots\\&&&&-1\end{pmatrix}\begin{pmatrix}1&&&&\\&0&&&\\&&0&&\\&&&\ddots&\\&&&&-1\\&&&&&0\end{pmatrix}\cdots\begin{pmatrix}1&&&\\&0&&\\&&-1&\\&&&\ddots\\&&&&0\end{pmatrix}\begin{pmatrix}1&&&\\&-1&&\\&&0&\\&&&\ddots\\&&&&0\end{pmatrix}$

5、$\because \dim V=\dim\operatorname{Im}\varphi+\dim\ker\varphi\leq\dim U+\dim\ker\varphi<\dim V+\dim\ker\varphi$

$\therefore \dim\ker\varphi>0$

6、根据U是子空间及线性映射的性质易证 $\varphi^{-1}(U)$ 是子空间，

定义 φ'：φ 在 $\varphi^{-1}(U)$ 上的限制，$\because 0\in U$ $\therefore \ker\varphi\in\varphi^{-1}(U)$

$\therefore \ker\varphi'=\ker\varphi\cap\varphi^{-1}(U)=\ker\varphi$

$\therefore \dim U+\dim\ker\varphi'=\dim U+\dim\ker\varphi'=\dim\varphi^{-1}(U)$

Recording start from here